Align Sample IDs Across Data Streams

Description

Computes the intersection of valid '(dataset_name, age)' pairs present in all three data sources: community (long format), abiotic (long format), and coordinate data. The returned table serves as the canonical sample index used by all subsequent data-preparation functions.

Usage

```
align_sample_ids(
  data_community_long = NULL,
  data_abiotic_long = NULL,
  data_coords = NULL,
  subset_age = NULL
)
```

Arguments

`data_community_long`	A data frame in long format containing at least the columns 'dataset_name' and 'age'.
`data_abiotic_long`	A data frame in long format containing at least the columns 'dataset_name' and 'age'.
`data_coords`	A data frame of spatial coordinates with 'dataset_name' stored as row names and columns 'coord_long' and 'coord_lat'.
`subset_age`	Optional numeric vector specifying age(s) to retain. If 'NULL' (default) all intersecting ages are kept.

Details

All three-way intersection logic is performed while the data remain in long (tidy) format, so no rowname parsing is needed. Passing a 'subset_age' value here — rather than inside a downstream modelling function — makes the subsetting explicit and cacheable as a pipeline target.

Value

A data frame with columns 'dataset_name' and 'age', containing only the '(dataset_name, age)' pairs present in all three inputs, arranged by 'dataset_name' then 'age'. When 'subset_age' is supplied the result is further filtered to those ages.

See Also

[prepare_community_for_fit()], [prepare_abiotic_for_fit()], [prepare_coords_for_fit()]